Contents

imud — Mathematical Reference (as implemented)

Document purpose. This is an audit reference for the *actual numerical methods executed by the imud source code*, at the release this file ships with. It is deliberately **not** a derivation of the methods imud "should" use, nor a restatement of the papers it draws on: every equation below was transcribed from the code and is annotated with the function, variable, and struct-field names so a reviewer can place each symbol against its implementation. Where the code approximates, deviates from, or specializes its cited source, an **As-implemented** note says so explicitly.

Typeset copy. `docs/math.pdf` in the source repository renders these equations properly for reading away from a terminal. It is committed but not packaged — it would only duplicate this file — and it is not linked from here, because a relative link to it would be dead in the installed doc tree. It is kept honest by `make check-math-pdf-stamp`, which compares it against this file by hash; after editing here, run `make math-pdf` and commit both. That needs `pandoc` and either `tectonic` (a single binary) or a XeLaTeX.

Intended reader. Someone auditing the mathematics itself — an estimation or geophysics specialist — who wants to check the equations and their data flow without reading C. Familiarity with quaternion attitude estimation, Kalman filtering, and spherical-harmonic geomagnetism is assumed.

Scope. All numeric routines in the daemon and its calibration tool: the MEKF attitude/heading filter, accelerometer/magnetometer measurement models, the heave and sea-state estimators, compass-health and engine-vibration metrics, timestamp anchoring, the offline magnetometer / Allan-variance / gyro-temperature calibration fits, and the World Magnetic Model evaluation. The WMM section states the *implementation shape* and cites the defining report rather than re-deriving spherical-harmonic theory.

Source-of-truth files. `src/fusion.c`, `include/fusion.h` (MEKF, heave, sea state); `src/imu.c` (live calibration application, Euler rates, rate of turn, timestamp anchor, engine detector, startup bias); `src/cal_math.c`, `include/cal_math.h` (offline fits, Allan variance); `src/wmm.c`, `include/wmm.h` (geomagnetic model); `src/cal.c` (calibration file I/O).

Citation convention. Each section ends with a **Source** line. Because imud implements standard methods rather than novel mathematics, most sources are the canonical reference for the technique. A citation marked *(code comment)* is the reference named in the source itself; one marked *(canonical)* is the standard literature reference for a technique the code does not cite in-line. See §15 References. Citations the maintainer may wish to verify against the exact edition are flagged in that section.

---

1. Conventions and reference frames

Frames.

- **Body frame** `b`: the sensor board axes after the optional mount rotation (§3). NED-compatible: X forward, Y starboard, Z down.
- **Navigation frame** `NED`: local North-East-Down.

Attitude quaternion. `mekf_t.q` $= q = [q_w, q_x, q_y, q_z]$ is a unit Hamilton quaternion, **scalar-first**, representing the **body→NED** rotation:

$$v_{NED} = R(q)\, v_b, \qquad R(q) \in SO(3).$$

The rotation matrix `q_to_R()` (`fusion.c:280`) is

$$R(q) = \begin{bmatrix} 1-2(q_y^2+q_z^2) & 2(q_xq_y - q_wq_z) & 2(q_xq_z + q_wq_y) \\ 2(q_xq_y + q_wq_z) & 1-2(q_x^2+q_z^2) & 2(q_yq_z - q_wq_x) \\ 2(q_xq_z - q_wq_y) & 2(q_yq_z + q_wq_x) & 1-2(q_x^2+q_y^2) \end{bmatrix}.$$

The Hamilton product `q_mul()` ($c = a \otimes b$, `fusion.c:243`) uses the standard scalar-first convention (Solà eq. 16).

Units. Gyroscope $\mathrm{rad} \cdot \mathrm{s}^{-1}$; accelerometer $\mathrm{m} \cdot \mathrm{s}^{-2}$; magnetometer µT on the wire, converted to **Gauss** ($\times 0.01$) inside the filter; angles rad; time s. Standard gravity `G_MS2` $= g = 9.80665\,\mathrm{m\,s^{-2}}$ (`fusion.c:43`).

Gravity reference. $g_{ref} = [0, 0, 1]$ (unit, NED, Z-down). The accelerometer's *specific force* at rest reads $-g$ on Z; the filter works with the **gravity direction** $z_a = -\hat{a}_b$ (§4.7).

As-implemented. MEKF state — quaternion, bias, wave acceleration, and the covariance P — is single precision (`float`); calibration fits and WMM are double precision. The quaternion is renormalized (`q_normalize`, `fusion.c:266`) after every multiplicative update, guarding unit-norm drift from float round-off, and P uses the Joseph form for the same reason (§4.5): the simple $(I - KH)P$ form slowly loses symmetry and positive-definiteness at 833 Hz over multi-day runs.

Where that split stops holding, and why the accumulators are double. The rule is not "filter state is float" but *nothing that accumulates over an unbounded number of steps is float.* Every leaky integrator and exponentially weighted statistic here advances by $\alpha\,\delta$ with $\alpha = \Delta t/\tau$, and that gain shrinks with the sample rate **and** with the time constant. Once α falls below half a float32 ULP of the state it is being added to, the update rounds to an exact no-op — not a gradual loss, a full stop — and $\varepsilon_{32} = 1.19 \times 10^{-7}$ is reachable from `imud.conf`: at 32 kHz a 1200 s sea-state window gives $\alpha = 2.6 \times 10^{-8}$.

Measured against a double-precision reference of the same arithmetic, both fed identical float-rounded inputs so only accumulator width differs:

```
rm -f test_fusion && make test_fusion CFLAGS="-D_GNU_SOURCE -O2 -Wall \
    -Wextra -std=c11 -pthread -Iinclude -DBENCH_SWEEP_PRECISION" && ./test_fusion
```

$\Delta t/\tau$	configuration	Hs error	T_z error
1.0×10^{-5}	833 Hz / 120 s **(default)**	0.000 %	0.000 %
2.1×10^{-7}	8 kHz / 600 s	0.128 %	0.491 %
5.2×10^{-8}	32 kHz / 600 s	1.823 %	0.982 %
2.6×10^{-8}	32 kHz / 1200 s	**17.995 %**	**6.865 %**

Heave, on its own τ axis, failed harder. Its leak is what bounds the drift of a double integration of accelerometer noise; when the leak becomes a no-op the estimator is a *plain* double integrator:

$\Delta t/\tau$	configuration	recovered amplitude error
2.6×10^{-6}	32 kHz / 12 s **(default)**	0.225 %
5.2×10^{-7}	32 kHz / 60 s	1.225 %
1.0×10^{-7}	32 kHz / 300 s	7.263 %
3.5×10^{-8}	32 kHz / 900 s	**6 543 975 %** — 253 m of "heave"

Both estimators' accumulators are therefore `double` (`seastate_t`'s six mean/variance pairs, `heave_t`'s `vel`/`disp`/`hp_y`), along with their gains and heave's gravity subtraction — a cancellation of 9.8 against 9.8 down to a signal three orders smaller. That moves the cliff to $\Delta t/\tau \sim 10^{-16}$, which no rate and window reach. Every cell above reads 0.000 % after. Inputs, outputs and the wire stay float: sensor resolution is coarser than float32 either way, so widening those would flatter the measurement without changing the hardware.

Independent confirmation from the default-on sweeps: `test_seastate_across_rates` was flat at 1.015 % Hs from 100 Hz to 16 kHz and ticked up to 1.046 % at 32 kHz. §4.7.2 recorded that endpoint as float32 accumulation. It is now 1.016 % — the plateau simply continues, and the uptick is gone.

One elapsed-time gain is worth naming separately. `mekf_ema_alpha` ($\alpha = 1 - e^{-\Delta t/\tau}$, exact for any feed rate) must be computed with `expm1f`: spelled `1.0f - expf(-x)` it subtracts two numbers differing by less than an ULP of 1.0, quantising the answer to multiples of 5.96×10^{-8} — 2.7 % error at 32 kHz against the 30 s health constant, with only 17 representable values.

Source: Solà 2017 *(code comment)*; Shuster 1993 for quaternion conventions *(canonical)*; Higham 2002 for the $1 - e^{-x}$ cancellation *(canonical)*. Measurements from `-DBENCH_SWEEP_PRECISION`.

---

2. Signal flow

One IMU sample traverses, in order:

1. **Driver read** → raw `imu_sample_t{accel[3], gyro[3], temp_c, chip_ts}` in board axes.
2. **Mount rotation** `apply_mount_rot_if_set()` (§3.1) → body axes.
3. **Calibration** `apply_imu_cal()` (§3.2): gyro temperature compensation, accel offset/scale. (Magnetometer: `apply_mag_cal()`, hard/soft iron.)
4. **MEKF** (§4): `mekf_predict` on every IMU sample; `mekf_update_accel` every sample; `mekf_update_mag` per magnetometer sample.
5. **Derived outputs**: Euler angles and heading (§4.10); Euler rates and rate of turn (§5); heave (§6); sea-state statistics (§7); compass-health (§8) and engine (§9) metrics.

Gyro **bias** is *not* removed in step 3; the MEKF estimates and subtracts it (§4.4). Timestamps are reconstructed from the chip counter (§11).

---

3. Calibration application (live path)

3.1 Mount rotation

`apply_mount_rot_if_set()` (`imu_math.c:393`) applies a fixed 3×3 board→body matrix R_{mount} = `cfg->mount_rot` in place, when `cfg->mount_set`:

$$v \leftarrow R_{mount}\, v, \qquad v \in \{a_b,\ \omega_b,\ m_b\}.$$

Applied identically to accel, gyro, and magnetometer vectors. Accumulated in double, stored back as float.

As-implemented. R_{mount} is supplied by configuration (a fixed installation rotation, e.g. yaw 180° for a stern-facing board); imud does not estimate it. It may be given as Euler angles (`rotation_euler_deg`), a named `preset`, or directly as a 3×3 `rotation_matrix`. The first two are orthonormal by construction; a directly-supplied matrix is validated at config load against both $R^\top R \approx I$ and $\det R > 0$ (rejecting reflections, which an orthogonality-only test would accept) and the daemon refuses to start if it fails.

3.2 Inertial calibration

`apply_imu_cal()` (`imu_math.c:26`), per axis i:

- **Gyro temperature compensation** (when `cal->has_gyro_temp`):

$$\omega_i \leftarrow \omega_i - c_i\,(T - T_{ref}),$$

  with c_i = `gyro_temp_coeff[i]`, T = `temp_c`, T_{ref} = `gyro_temp_ref_c` (fit in §12.6).
- **Accel offset/scale** (when `cal->has_accel`):

$$a_i \leftarrow (a_i - o_i)\,s_i,$$

  with o_i = `accel_offset[i]`, s_i = `accel_scale[i]`.

3.3 Magnetometer hard/soft-iron

`apply_mag_cal()` (`imu_math.c:44`), when `cal->has_mag`:

$$m_{cal} = S_{soft}\,(m_{raw} - b_{hard}),$$

with b_{hard} = `mag_hard_iron[3]`, S_{soft} = `mag_soft_iron[3][3]`. The correction parameters are produced offline (§12.2–12.3).

As-implemented. The accel model is diagonal scale + offset (no off-diagonal cross-axis / misalignment terms). The magnetometer model is a full 3×3 soft-iron matrix times a hard-iron offset, but that matrix is fitted from a **2-D** swing (§12.3), so its out-of-plane (dip) accuracy is limited — which is exactly why the filter defaults to heading-only magnetometer fusion (§4.8).

Source: standard sensor error model; Titterton & Weston 2004 §12 *(canonical)*.

---

4. Multiplicative Extended Kalman Filter

The attitude/heading core is an **error-state (multiplicative) EKF**. The nominal state carries the full quaternion and gyro bias; the Kalman filter operates on a minimal 6-D **error state**, and corrections are injected multiplicatively so the quaternion never leaves the unit sphere.

4.1 State and covariance

Nominal state (`mekf_t`, `fusion.h:46`):

- q — unit quaternion, body→NED (`q[4]`).
- b — gyro bias, $\mathrm{rad}\cdot\mathrm{s}^{-1}$ (`bias[3]`).
- a_w — wave acceleration, body frame, normalized gravity units (`wave_acc[3]`); see §4.1.1.

Error state (never stored explicitly; the axis of `P`):

$$\delta x = [\,\delta\theta\ (3)\ \mid\ \delta b\ (3)\ \mid\ \delta a_w\ (3)\,] \in \mathbb{R}^9,$$

$\delta\theta$ = small-angle rotation error (rad), δb = bias error, δa_w = wave-acceleration error. Covariance $P \in \mathbb{R}^{9\times 9}$ (`P[MEKF_N][MEKF_N]`, `MEKF_N = 9`), symmetric PD; top-left 3×3 is attitude-error covariance (rad²), the middle 3×3 is bias-error covariance ($(\mathrm{rad}\cdot\mathrm{s}^{-1})^2$), the trailing 3×3 is wave-acceleration-error covariance (g²).

The wave block is **appended**, not inserted, so the attitude and bias blocks keep their indices: `mekf_get_state()` reads `P[0:3][0:3]` and `P[3][3]`, `P[4][4]`, `P[5][5]` exactly as before and the wire format is unchanged.

4.1.1 The wave-acceleration state The gravity measurement is contaminated in a seaway by wave-orbital acceleration, which is **correlated over ~0.5–1.5 s, not white**. A white R_a therefore tells the filter that 833 samples per second are 833 independent measurements of gravity when they are closer to one per wave. The consequence is not subtle: the filter's covariance collapses, its gain vanishes, and it diverges into its own over-confidence (measured: attitude RMS 9.5°/11.4°, NIS ≈ 56 — §4.7).

a_w is modelled as a **first-order Gauss–Markov (Ornstein–Uhlenbeck) process** with steady-state standard deviation σ and correlation time τ:

$$\dot{a}_w = -\frac{1}{\tau} a_w + w, \qquad \mathbb{E}[a_w a_w^{\mathsf{T}}] = \sigma^2 I_3.$$

σ is configured in $\mathrm{m}\cdot\mathrm{s}^{-2}$ (`mekf_wave_accel`, the units a spec sheet and `imud-cal fit-ra` both speak) and stored as the variance $\sigma_g^2 = (\sigma/g)^2$ in `wave_sig2`; τ is `mekf_wave_accel_tau_s`. Either at 0 disables the state, and the disabled filter is bit-for-bit the pre-1.7 6-state filter — P's wave rows and columns stay identically zero, Φ's block is I, and the measurement Jacobian's block is never executed.

Frame. a_w is carried in the **body** frame. Wave-orbital acceleration is arguably a property of the seaway and so NED-stationary, which would add a transport term $\Phi_w = e^{-dt/\tau}(I - [\omega\, dt]_\times)$. That was implemented and measured (`-DWAVE_TRANSPORT` in `fusion.c`): it changes the benchmark in the third decimal and is slightly *worse* on yaw-only attitude RMS (2.308° → 2.325°), because at ±15° of roll the frame rotation is small compared with τ. The body-frame form is kept for the simpler Jacobian.

Observability. $\delta\theta$ and δa_w both act in the tangent plane of the measured direction, so they are separated *only* by their dynamics — attitude error is a random walk driven by gyro noise, wave acceleration decays with τ. Too large a σ or τ lets the wave state absorb genuine tilt error; §4.7 records the sweep that bounds this.

The nominal/error relationship is the **right** (local) perturbation $q_{true} = \hat{q} \otimes \delta q(\delta\theta)$, established by the sign of the measurement Jacobian in §4.5.

Source: Solà 2017 §5.4 *(code comment)*; Markley & Crassidis 2014 §6.1; Trawny & Roumeliotis 2005 *(canonical).*

4.2 Initialization — `mekf_init()` (`fusion.c:1019`)

Nominal: $q = [1, 0, 0, 0]$, b = `gyro_bias_init` (from the startup still window, §10; may be 0).

Initial covariance (diagonal):

$$P_0[0{:}3] = (0.175)^2\ \mathrm{rad}^2\ (\approx 10°), \qquad P_0[3{:}6] = (0.001)^2\ (\mathrm{rad\,s^{-1}})^2, \qquad P_0[6{:}9] = \sigma_g^2.$$

The wave block starts at its **steady state**, not at a wide acquisition value: the Gauss–Markov process is stationary, so there is no transient to model. (It is seeded to 0 when the state is disabled, which is what keeps the block inert.)

Discrete process-noise variances, step $dt = 1/\mathrm{ODR}$:

$$Q_g = N_g^2\, dt, \qquad Q_b = N_b^2\, dt,$$

with N_g = `mekf_gyro_noise`, N_b = `mekf_gyro_bias` (datasheet noise densities). Stored as `f->Qg`, `f->Qb`.

Measurement-noise variances:

$$R_a = \left(\frac{N_a}{g}\right)^2 \text{ODR}, \qquad R_m = N_m^2\, f_{s,\text{mag}},$$

`f->Ra` (`mekf_derive_tuning`, `fusion.c:897`) in normalized (gravity-direction) units, `f->Rm` (`:898`) in Gauss², with N_a = `mekf_accel_noise`, N_m = `mekf_mag_noise`, $f_{s,\text{mag}}$ = `mag_odr_hz`.

Derived thresholds: accel skip band $[\,1 - s_k,\ 1 + s_k\,]$ with s_k = `accel_skip_thresh`; mag_reject_sq = (mag_reject_gauss)^2; convergence threshold $conv\_thresh = 3\,\theta_c^2$ with $\theta_c = \max(0.5°,\ 0.30 \arcsin \sigma_g)$ — 1.40° at the default σ. It scales with σ_g rather than being flat because with the wave state P reports a believable steady-state attitude variance, and that has a floor near 0.9°/axis: a fixed 0.5° threshold would never be reached (§4.1.1). m_ref EMA gain $mref\_alpha = 1/(\tau_{mref}\, f_{s,\text{mag}})$ with $\tau_{mref} = 300$ s (`:941`).

As-implemented. Q_g, Q_b are *per nominal step*; during prediction they are rescaled by the actual dt (§4.4). The bias process is modeled as a random walk whose density N_b is a **tuning constant deliberately held above** the measured in-run bias instability (see `docs/capture.md`); it is not driven by the Allan-variance characterization of §12.5.

Source: Solà 2017 §4.1 for discrete process noise *(code comment)*.

4.3 Alignment — `mekf_align()` (`fusion.c:1077`)

Deterministic initial attitude from one static accel+mag pair (a tilt-then-heading decomposition, TRIAD-family).

The caller supplies a window mean, not one sample (`imu.c`, the align loop): the accelerometer and magnetometer are averaged over `align_window_sec` before this is called. That window matters far more than its obscurity suggests, because whatever tilt error survives it is baked permanently into m_{ref}'s dip (§4.8.1) and, in 3-D mode, becomes a constant roll/pitch bias. One second — the hardcoded value before 1.7 — is about a fifth of a typical roll period, so it averages an arbitrary fraction of the cycle. Measured over the 12-seed wave benchmark, attitude RMS:

window	yaw-only (default)	3-D	3-D NEES(strict)
1 s	47.7°	6.72°	339
2 s	2.28°	5.78°	250
3 s	2.11°	2.38°	38.9
5 s (default)	**2.19°**	**1.18°**	**5.74**
15 s	2.19°	0.90°	1.71
30 s	2.24°	0.93°	1.94

The marine default is flat from ~2 s; 3-D keeps improving to ~15 s. The default is 5 s, and the cost of a longer window is purely startup latency before the filter produces usable attitude — worth paying at a mooring, not underway.

Tilt from accelerometer. With $\hat{g}_b = -a_b/\|a_b\|$ the gravity direction in body (guard: reject if $\|a_b\| < 0.5g$):

$$\phi = \text{atan2}(\hat{g}_{b,y},\ \hat{g}_{b,z}), \qquad \theta = \text{atan2}(-\hat{g}_{b,x},\ \sqrt{\hat{g}_{b,y}^2 + \hat{g}_{b,z}^2}).$$

Tilt quaternion $q_{tilt} = q_\theta \otimes q_\phi$ built directly from half-angles (`mekf_align`, `fusion.c:1094`).

Heading from magnetometer. Rotate m_b by $R(q_{tilt})$ into the tilt-levelled frame; with horizontal components m_x, m_y there,

$$\psi = \text{atan2}(-m_y,\ m_x).$$

Full attitude $q = q_\psi \otimes q_{tilt}$, $q_\psi = [\cos\frac{\psi}{2}, 0, 0, \sin\frac{\psi}{2}]$.

Magnetic reference. On first alignment, `m_ref` (NED, Gauss) is set from the measurement rotated to NED and scaled µT→Gauss:

$$m_{ref} = 0.01\,R(q)\,m_b.$$

As-implemented / audit note. The heading sign is $\operatorname{atan2}(-m_y, m_x)$, **not** $\operatorname{atan2}(+m_y, m_x)$; the levelled field is the true NED field rotated by $-\psi$, so the negative sign recovers vessel yaw. (The `+m_y` form returns $-\psi$, mirrored about north — a historical bug invisible when aligning while pointing north; see the in-code comment at `fusion.c:1104`.) The initial `m_ref` inherits any alignment tilt error in its magnitude/dip, which the quiescence-gated EMA (§4.8) and the WMM invariants (§4.9) later remove; its horizontal **direction** is the heading datum and is never subsequently adapted.

Source: tilt/heading coarse alignment — Farrell 2008 §10; TRIAD lineage Shuster & Oh 1981 *(canonical)*.

4.4 Prediction — `mekf_predict()` (`fusion.c:1134`)

Per IMU sample, with measured interval dt (from hardware timestamps, §11; falls back to nominal 1/ODR).

Bias-corrected rate: $\omega = s.\text{gyro} - b$.

Quaternion propagation by the exponential map (`q_from_rotvec`, `fusion.c:293`):

$$q \leftarrow q \otimes \exp\left(\tfrac{1}{2}\,[0,\ \omega\,dt\,]\right), \qquad \exp(\phi) = \left[\cos\tfrac{|\vartheta|}{2},\ \tfrac{\sin(|\vartheta|/2)}{|\vartheta|}\,\vartheta\right],$$

$\vartheta = \omega\,dt$, with the small-angle branch $\delta q \approx [1, \frac{1}{2}\vartheta]$ for $|\vartheta| < 10^{-7}$. Renormalized after.

Covariance propagation $P \leftarrow \Phi P \Phi^\top + Q_d$, first-order discrete transition (`mekf_predict`, `fusion.c:1162`):

$$\Phi = \begin{bmatrix} I_3 - [\omega]_\times\,dt & -I_3\,dt & 0_3 \\ 0_3 & I_3 & 0_3 \\ 0_3 & 0_3 & \varphi\,I_3 \end{bmatrix}, \qquad Q_d = \operatorname{diag}(Q_g \tfrac{dt}{dt_0} I_3,\ Q_b \tfrac{dt}{dt_0} I_3,\ Q_w I_3),$$

$[\omega]_\times$ the skew-symmetric cross-product matrix; the gyro/bias noise is rescaled by dt/dt_0 (dt_0 = nominal step) so variance grows linearly with the real interval. P is symmetrized after. Convergence flag $\operatorname{tr}(P[0{:}3]) < conv\_thresh$.

The wave block is discretized exactly, not to first order:

$$\varphi = e^{-dt/\tau}, \qquad Q_w = \sigma_g^2\,(1 - \varphi^2), \qquad \hat{a}_w \leftarrow \varphi\,\hat{a}_w.$$

This is not fastidiousness. τ is of order 0.5 s while the $|a|$ skip band routinely leaves gaps of the same order, so dt/τ is **not** small on exactly the steps that matter, and a first-order expansion would both over-decay the state and mis-size its noise while bridging a gap. The exact form is stationary for any dt: feed it a step of 10τ and it returns the state to zero with variance σ_g^2, which is the correct answer — so a scheduling hiccup cannot destabilize the filter. It is also why Q_w is *not* put through the dt/dt_0 rescaling: it is already exact for this dt.

With the state disabled $\varphi = 1$ and $Q_w = 0$, leaving an identity block that touches nothing.

As-implemented. Prediction uses the plain $\Phi P \Phi^\top + Q$ form (not Joseph — the Joseph stabilization is applied on the *updates*, §4.5). Φ is a first-order (zero-hold) approximation of $\exp(F_c\,dt)$; valid because $\|\omega\| dt \ll 1$ at supported ODRs. The $[\omega]_\times$ sign in $\Phi_{[0:3,0:3]}$ is that of the right-perturbation error convention.

Source: Solà 2017 eq. 259 (integration), eq. 268 (covariance) *(code comment)*.

4.5 Generic vector measurement update — `eskf_update()` (`fusion.c:495`)

Measurement of a known NED reference observed in body: predicted h, actual z, isotropic noise R_{noise}, gate `chi2_gate`.

Jacobian (3×9), attitude block only (`dir_jacobian`):

$$H = \big[\, [h]_\times \;\big|\; 0_3 \;\big|\; 0_3 \,\big], \qquad [h]_\times = \begin{bmatrix} 0 & -h_2 & h_1 \\ h_2 & 0 & -h_0 \\ -h_1 & h_0 & 0 \end{bmatrix}.$$

The **gravity** update uses a different Jacobian when the wave state is enabled — see §4.5.1. The mag channels use this one, unchanged.

Innovation covariance and gain:

$$S = HPH^\top + R_{noise} I_3, \qquad K = PH^\top S^{-1},$$

S^{-1} by Cramer's rule (`m33_inv`, `fusion.c:95`; singular ⇒ skip).

The singularity test is relative, not absolute (1.7). S carries physical units, and for the gravity update they are tiny: with $R_a \approx 4.2 \times 10^{-5}$ and the attitude block converged to ~0.4°, $\det S \approx R_a \lambda^2 \approx 6 \times 10^{-13}$ at a condition number of about 3. The previous absolute test $|\det S| < 10^{-12}$ therefore declared a well-conditioned matrix singular and made `eskf_update` return -1, **silently dropping 87% of accel updates in the wave benchmark** and pinning attitude uncertainty at whatever value made $\det S \approx 10^{-12}$ rather than letting it converge. `m33_inv` now compares $|\det|$ against $10^{-6}\,\|A\|_{\max}^3$, which is unit-free, tests rank deficiency (what "singular" means), and still sits well above float's $\approx 1.2 \times 10^{-7}$ epsilon. §4.7 records what this exposed.

Innovation $\nu = z - \hat{z}$ ($\hat{z} = h$ here), with **robust (Huber-style) gating** on the Mahalanobis distance $d^2 = \nu^\top S^{-1} \nu$:

$$\begin{cases} \text{reject the update} & d^2 > 9\,\gamma \\ \nu \leftarrow \nu\,\sqrt{\gamma/d^2} & \gamma < d^2 \le 9\gamma \\ \nu \text{ unchanged} & d^2 \le \gamma \end{cases}$$

$\gamma =$ `chi2_gate`. The middle branch caps the innovation's influence at the gate distance rather than discarding it.

Correction and injection:

$$\delta x = K\,\nu, \qquad q \leftarrow q \otimes \exp(\delta\theta), \qquad b \leftarrow b + \delta b, \qquad a_w \leftarrow a_w + \delta a_w,$$

$\delta\theta = \delta x[0{:}3]$, $\delta b = \delta x[3{:}6]$, $\delta a_w = \delta x[6{:}9]$; quaternion renormalized. The a_w injection is unconditional: with the state disabled P's wave rows are zero, so K's are too and it adds exactly $0.0f$. The *mag* channels inject it as well — their H has no wave block, but the cross-covariance the accel path builds up means K does, and that is correct Kalman book-keeping.

Covariance — Joseph form, with error-state reset folded in:

$$P \leftarrow \big[G(I - KH)\big]\, P\, \big[G(I - KH)\big]^\top + R_{noise}\,(GK)(GK)^\top,$$

then symmetrized. (Isotropic $R \Rightarrow KRK^\top = R_{noise} KK^\top$.)

Error-state reset. Injecting $\delta\theta$ into the quaternion moves the linearisation point, so P is rotated by the reset Jacobian $G = I - \frac{1}{2}[\delta\theta]_\times$ (Solà eq. 285). Only the attitude block resets — the gyro-bias and wave-acceleration errors are additive and carry over — so the applied transform is $G_{full} = \mathrm{diag}(G, I_3, I_3)$, implemented by premultiplying the first three rows of K and $(I - KH)$. Cost is ~81 multiply-adds against the Joseph block's ~450. Measured over the 12-seed wave benchmark this improves yaw-only attitude RMS 4.10° → 3.57° and covariance consistency (NEES 32.4 → 28.4), with 3-D attitude neutral (+2%).

Huber cap. The cap replaces a hard χ^2 reject so that wave-orbital acceleration (which swings gravity *direction* while keeping $|a| \approx g$) deweights rather than starves the filter; only gross outliers ($d^2 > 25\gamma$, `GROSS_REJECT_MULT`) are rejected. Because S contains P, the cap is naturally inactive during acquisition (large P) and engages only once confident — no explicit convergence gate needed.

Gross-reject threshold (1.7: $9\gamma \to 25\gamma$). The original 9γ was too tight to be a fault gate. Over the 12-seed benchmark it discarded 26% of 3-D accel updates — routine wave motion, not outliers — which starved the filter, and the resulting drift produced larger innovations that tripped the gate more often still. Widening it (3-D / yaw-only pairs):

reject	3-D att	yaw att	NEES(tr) 3-D/yaw	NIS 3-D/yaw	reject rate
9γ	6.85°	3.57°	28.9 / 21.1	30.7 / 27.3	.262 / .043
16γ	5.45°	2.32°	16.7 / 7.9	22.3 / 25.2	.012 / .000
25γ	5.65°	2.31°	18.3 / 7.8	19.3 / 25.2	.007 / .000
64γ	5.62°	2.31°	18.3 / 7.8	16.7 / 25.2	.000 / .000

Attitude RMS improves 17% (3-D) and 35% (yaw-only), NEES roughly halves, and the worst-draw spread collapses (3-D 13.7°→7.0°, yaw 9.6°→2.3°) — the spread §10.8 had attributed to scenario luck. The benefit saturates by 25γ, which keeps a genuine fault gate at 5× the cap radius instead of 3×.

Known inconsistency (measured, deliberately retained). The cap scales ν but leaves K — and hence the covariance update — at full confidence, so P contracts as though the measurement had been fully trusted even when the correction was attenuated up to 3×. Making this consistent was tried three ways and measured over the 12-seed wave benchmark, where NEES ≈ 1 would indicate a self-consistent filter:

Re-measured after the 25γ change above (the earlier figures were taken while the reject gate was corrupting every run). Rebuild with `-DHUBER_VARIANT=n`:

n	variant	3-D att	yaw att	NEES(tr) 3-D/yaw	NIS 3-D/yaw
0	ν-capping (shipped)	5.65°	2.31°	18.3 / 7.8	19.3 / 25.2
1	$K \leftarrow wK$ (exact Joseph, suboptimal gain)	8.85°	3.09°	39.9 / 12.4	17.0 / 28.5
2	$R \to R/w$ (IRLS inflation)	7.79°	5.57°	31.5 / 40.5	15.1 / 43.3
3	$R \to R/w^2$	8.48°	4.74°	35.0 / 27.9	18.2 / 36.3

Every variant is **both** less accurate and less self-consistent — by a wider margin than before. Inflating R is additionally wrong on its own terms because $P \gg R$ here, a regime where $K = P/(P+R)$ is near unity and $R \to R/w$ barely moves it, losing the bounded per-sample influence the seaway robustness depends on.

The reason these fail is **not** that R_a is mistuned. R_a *is* over-optimistic (NIS ≈ 20, §4.7), but retuning it does not rescue the family — they stay worse across the whole sweep. The cap is a *robustness* device, not a statistical one: contracting P by the attenuated gain faithfully records having learned less from that sample, but the consequence is a larger P, hence a larger gain on the **following** samples — which in a seaway carry the same wave contamination. The inconsistency does useful work, holding the gain down exactly when measurements are least trustworthy. `innov_weight` / `innov_reject` on the wire (§8.1) expose how hard the cap is being leaned on; `nis_accel` / `nis_mag` (§8.2) expose whether the model under it is right.

Source: EKF update Kalman 1960; Joseph form Bucy & Joseph 1968; robust innovation capping Huber 1964; Mahalanobis gating Bar-Shalom et al. 2001 *(canonical)*. Jacobian sign Solà 2017 §7 *(code comment)*.

4.5.1 Wave-aware gravity Jacobian — `wave_jacobian()` When the Gauss–Markov state is enabled, the gravity update no longer predicts h; it predicts the *contaminated* direction the accelerometer actually sees. With the specific force written in normalized gravity units,

$$v \equiv h - \hat{a}_w, \qquad \hat{z} = \frac{v}{\|v\|}, \qquad \nu = z - \hat{z}.$$

Perturbing both contributions ($h_{true} = \hat{h} + [\hat{h}]_\times \delta\theta$, $a_{true} = \hat{a}_w + \delta a_w$) gives $\delta v = [\hat{h}]_\times \delta\theta - \delta a_w$, and differentiating the normalization introduces the **tangent projector** $P_{\hat{z}} = I - \hat{z}\hat{z}^\mathsf{T}$:

$$\delta\hat{z} = \frac{P_{\hat{z}}}{\|v\|}\,\delta v \quad\Longrightarrow\quad H = \left[\ \tfrac{1}{\|v\|} P_{\hat{z}} [h]_\times \ \middle|\ 0_3 \ \middle|\ -\tfrac{1}{\|v\|} P_{\hat{z}}\ \right].$$

The projector is load-bearing, not tidiness. It is what makes $\hat{z}^\mathsf{T} H = 0$ hold *exactly*, since $P_{\hat{z}}\hat{z} = 0$. That in turn makes $\hat{z}$ an eigenvector of S with eigenvalue R, which is the sole premise of the $\mathbb{E}[d^2] = 2$ derivation in §8.2 — i.e. of the `nis_accel` wire field meaning "1.0 = consistent". Writing the Jacobian the way a first pass naturally would, $[\hat{h}]_\times/\|v\|$ and $-I/\|v\|$, leaves $\hat{z}^\mathsf{T} H = O(\|\hat{a}_w\|)$; the radial component then leaks into d^2 and NIS is biased. Measured: the unprojected form moves the ground-truth benchmark NIS from 0.99 to 0.76 while leaving every other test in the suite passing (`test_wave_gm_ground_truth`).

At $\hat{a}_w = 0$ this reduces algebraically to `dir_jacobian`: $v = h$ is already a unit vector and $(I - hh^\mathsf{T})[h]_\times = [h]_\times$ because $h^\mathsf{T}[h]_\times = 0$.

$\|v\|$ is guarded at 0.5; below that the acceleration is comparable to gravity itself and the linearisation is meaningless, so the update falls back to the plain direction form.

4.6 Scalar heading update — `eskf_update_yaw() (fusion.c:758)`

Heading-only correction (a 1-D measurement). Innovation y (rad, wrapped to $\pm\pi$), variance R_{noise}. The Jacobian projects the error onto the NED-down axis expressed in body:

$$H = [\ -R[2][:]\ \mid\ 0_3\ ] \in \mathbb{R}^{1\times 6},$$

$R[2][:]$ = third row of $R(q)$. Then

$$S = HPH^\top + R_{noise}, \qquad K = PH^\top / S,$$

with the same Huber policy on $d^2 = y^2/S$ against $\gamma_\psi = 6.63$ (χ^2_1 99%), rank-1 Joseph covariance update.

Source: scalar-measurement EKF specialization of §4.5 *(canonical)*.

4.7 Accelerometer update — `mekf_update_accel() (fusion.c:1374)`

Speed-aided centripetal correction. In a turn the accelerometer senses $a = a_{platform} - g$, and with speed-over-ground v (body $\approx [v, 0, 0]$, leeway neglected) the centripetal term is $\omega \times v = [0,\ \omega_z v,\ -\omega_y v]$. When `speed_mps` > 0.1:

$$a_y \leftarrow a_y - \omega_z\, v, \qquad a_z \leftarrow a_z + \omega_y\, v,$$

$\omega_{y,z}$ bias-corrected. `speed_mps=0` (no GPS) $\Rightarrow$ no-op.

Quiescence EMA (updated every sample, incl. gated), $\tau \approx 2$ s:

$$q_{quiet} \leftarrow q_{quiet} + ((a_g - 1)^2 - q_{quiet})\,\tfrac{dt}{2}, \qquad a_g = \|a\|/g.$$

Skip gate. If $a_g \notin$ [`accel_skip_lo`, `accel_skip_hi`], skip (linear acceleration).

Measurement. Gravity direction $z = -a/\|a\|$; prediction $h = R^\top g_{ref} = R[2][:]$ (third row). Effective noise $R_{eff} = R_a\cdot$ `Ra_scale`. Then `eskf_update(h, z, R_eff, 11.34)` with $\gamma = 11.34$ (χ^2_3 99%). The gravity direction z is stashed as `g_body` (attitude-independent dip reference for §4.8).

Protected axis. h is also passed as `eskf_update`'s `protect` argument, which projects the attitude rows of K onto the plane orthogonal to it:

$$K_{\delta\theta} \leftarrow (I - hh^\top)\, K_{\delta\theta}.$$

Two properties follow, and they are what the update is entitled to claim:

$$\delta\theta \cdot h = 0, \qquad h^\top P^+ h = h^\top P^- h,$$

the second because $K^\top(h, 0, 0) = (h^\top K_{\delta\theta})^\top = 0$ leaves both $(I - KH)^\top$ and $KRK^\top$ inert along h. Joseph form holds for any gain, so nothing else in §4.5 changes.

$H_{\delta\theta} h = 0$ already holds exactly for both Jacobians — the projection subtracts nothing at small angle. It exists for the large-angle regime. Once $\operatorname{tr} P[0{:}3]$ reaches $O(1)$ rad², which is where an unaided yaw axis ends up, the gain is large enough that a single innovation moves q further than the first-order reset $G = I - \frac{1}{2}[\delta\theta]_\times$ tracks h; the misalignment leaks the unobservable variance into the observable subspace, the next update annihilates it, and the filter reports a collapsed covariance and a heading correction gravity cannot justify.

Attitude rows only. A gyro-bias component along h is equally unobservable from this measurement at this instant, but h rotates in the body frame as the platform moves, so that component becomes observable a moment later. A yaw error does not: Φ rotates $\delta\theta$ with the body and h rotates with the body, so $\delta\theta \parallel h$ stays parallel to h for all time. Projecting rows 0–2 is therefore correct and projecting rows 3–5 would not be. The magnetometer passes `protect = NULL` (§4.8): it is the measurement that does carry heading.

As-implemented. `Ra_scale` is set to 4 by the fusion thread while engine vibration is detected (§9) — vibration is high-frequency, near-zero-mean, so deweighting (not gating) is correct. Wave *direction* disturbance is handled by the Huber cap (§4.5), **not** by magnitude-based deweighting, which benchmarking showed rectifies wave-phase-correlated error into an attitude/bias offset. The centripetal model assumes zero leeway and no vertical velocity.

Health-EMA gain (1.7 fix). `innov_weight`, `innov_reject` and `nis_accel` are fed only by samples that clear the skip gate. In a seaway that gate discards 85–95% of samples, so a per-update gain sized from the IMU ODR (`gate_alpha` $= 1/(\tau f_{odr})$) stretched the intended $\tau = 30$ s to ten minutes or more, and the metrics reported their seed values rather than the data. The accel path now derives the gain from elapsed time instead:

$$\alpha = 1 - \exp(-\Delta t/\tau),$$

Δt accumulated over *every* sample, accepted or skipped. This is exact at any feed rate and reduces to $\Delta t/\tau$ when updates are frequent.

What R_a can and cannot do. R_a derives from the datasheet noise floor, and the filter is measurably over-confident about it: mean NIS over the wave benchmark is 19 (3-D) / 25 (yaw-only) where a consistent model reads 1. The natural remedy — raise `mekf_accel_noise` until NIS ≈ 1 — does not work. Sweeping it (12 seeds, both mag modes):

N_a	3-D att	yaw att	NEES(tr) 3-D/yaw	NIS 3-D/yaw
0.0022 (default)	5.65°	2.31°	18.3 / 7.8	19.3 / 25.2
0.004	7.93°	11.13°	114 / 304	33.7 / 39.6
0.008	7.57°	11.09°	102 / 285	10.6 / 13.5
0.015	6.98°	10.29°	81 / 241	3.2 / 4.2
0.030	5.74°	**8.58°**	50 / 156	**0.82 / 1.03**
0.050	5.07°	7.01°	34 / 87	0.32 / 0.34
0.120	4.72°	4.33°	17 / 17	0.06 / 0.05
0.300	4.47°	2.48°	6.4 / 2.2	0.01 / 0.01

Three things to read off it:

1. **NIS ≈ 1 costs accuracy.** $N_a = 0.03$ makes the innovations statistically consistent, but the marine (yaw-only) default degrades 2.31° → 8.58° and its NEES rises 7.8 → 156. Weakening the gravity correction makes the filter lean on gyro integration, so the *state* error grows even as the *innovations* start to look honest.
2. **The two consistency criteria disagree.** NIS falls monotonically with N_a while NEES(trace) has its minimum near the default. No single scalar satisfies both.
3. **The inconsistency is structural, not a scale error.** NEES(strict) for 3-D stays pinned in the 44–64 band across the entire sweep — four orders of magnitude of R_a — so P's *shape* is wrong, and no isotropic scalar can fix that.

The cause is that the seaway residual is wave-orbital: correlated with wave phase, with a measured correlation time of order a second (`imud-cal fit-ra`), not white. A white isotropic R cannot describe a coloured disturbance; making its *variance* right necessarily gets its *spectrum* wrong. The real fix is to model the correlation, which is what §4.7.1 does.

The default is therefore unchanged, and is additionally a sharp local optimum: $N_a \in [0.003, 0.03]$ is markedly **worse** than either side of it, so hand- tuning it upward "a little" lands in the worst available region.

> **Caveat on the table above.** These rows were measured before the `m33_inv` singularity bug of §4.5 was found, i.e. through a filter that was discarding 87% of its accel updates. The *conclusions* survive — no scalar R_a fixes a coloured disturbance, and the 44–64 NEES(strict) band really was structural — but the absolute numbers are not reproducible on current code and the table is kept as the historical record of why §10.5 was undertaken.

4.7.1 The Gauss–Markov wave state — outcome Fixing `m33_inv` removed an accidental 87% decimation of accel updates. That decimation had been doing real work: it crudely decorrelated the wave-contaminated samples, which is why the filter looked as good as it did. With every sample fed to a white R, the filter believes it has 833 independent gravity measurements per second, P collapses, and it diverges. Modelling the disturbance as a first-order Gauss–Markov process (§4.1.1) makes repeated correlated samples correctly stop adding information. Over the 12-seed wave benchmark:

	3-D att	3-D hdg	yaw att	yaw hdg	NEES(tr) 3-D/yaw	NEES(st) 3-D/yaw	NIS 3-
old baseline (bug present)	5.653°	3.065°	2.309°	1.961°	18.3 / 7.8	57 / 38	19.3 /
`m33_inv` fixed, no wave state	9.488°	7.255°	11.424°	8.801°	259 / 252	933 / 1086	56.5 /
+ wave state (shipped)	**4.452°**	**0.828°**	**2.308°**	**1.016°**	**3.47 / 0.99**	335 / **10.1**	**1.01 /**

(For 3-D attitude and NEES(strict), §4.8.1 carries the authoritative figures: it measures them with the benchmark's alignment corrected. The wave-state conclusion here rests on the yaw-only default and on the NIS column.)

Against the old baseline: 3-D attitude −21%, 3-D heading −73%, yaw heading −48%, yaw attitude a dead heat, NIS from 19–25 to $\approx$1, NEES(trace) from 18.3/7.8 to 3.47/0.99, and both the Huber cap and the gross-reject gate go completely idle (weight 1.000, reject 0.0000) — the filter no longer needs robustness machinery to survive an ordinary seaway.

Tuning, and why it is not curve-fitting. σ and τ were chosen on a *broadband* scenario (`SCEN_GM` in `test_fusion.c`) whose disturbance is a genuine Gauss–Markov process of known σ and τ, because a single tone has no correlation time and fitting τ to one is fitting the benchmark. The tone is the held-out validation. Three independent lines agree on $\sigma \approx 0.8\ \mathrm{m \cdot s^{-2}}$:

- it is the broadband scenario's ground truth, at which the filter reports NIS = 0.99 (`test_wave_gm_ground_truth`);
- it is exactly the tone scenario's true per-axis RMS ($1.2/\sqrt{2}$);
- `imud-cal fit-ra` independently recovers $0.67\ \mathrm{m \cdot s^{-2}}$ from a capture of that tone, seen only through a file, a replayed filter and a 67%-decimating skip band.

$\tau = 0.5$ s sits inside the range fit-ra measures. Neither knob is a free parameter. The full grid is reproducible with

```
rm -f test_fusion && make test_fusion CFLAGS="-D_GNU_SOURCE -O2 -Wall \
    -Wextra -std=c11 -pthread -Iinclude -DBENCH_SWEEP_WAVE" && ./test_fusion
```

The degeneracy ridge. $\delta\theta$ and δa_w are separated only by their dynamics, so an over-large τ lets the wave state absorb real tilt. The sweep shows exactly that: at $\sigma = 0.9$ the yaw-only attitude RMS runs 2.17° / 2.29° / 2.40° / 2.72° for $\tau = 0.3$ / 0.5 / 0.6 / 0.8 s, and by $\tau = 2$ s it has collapsed to 7.14°. Short τ is the safe side; the shipped 0.5 s sits at the knee. A visible symptom of straying too far is `bias_z` drifting, which the benchmark bounds independently.

The column that looked worse: 3-D NEES(strict), 57 → 335. Two things were going on, and the first attribution of this to "the swing-circle mag calibration is structurally 2-D" was **wrong** — the benchmark synthesises its magnetometer data from the true attitude with nothing but white noise, so no calibration defect can be what it measures. See §4.8.1: it is the magnetic reference's DIP error, and ~96% of it was the benchmark aligning from a single instantaneous sample where the daemon averages a window.

4.7.2 The filter across the whole rate ladder Every figure above, and every figure elsewhere in this document, was measured at one point: 833 Hz IMU with a ~104 Hz magnetometer. The drivers advertise **29 IMU rates from 12 Hz to 32 kHz and 13 magnetometer rates from 1 Hz to 1 kHz**, and each of those 377 pairings is a configuration `imud.conf` will accept. This section is what the other 376 do.

Two instruments, because the questions are different sizes. `test_rate_derivations` walks all 377 pairings on every build, asserting the derived tuning is correct and the filter stays numerically sane — it deliberately asserts nothing about accuracy, since $R_a \propto f_{odr}$ retunes the filter at every rung and the 833 Hz bounds are meaningless elsewhere. Accuracy comes from an opt-in sweep along two axes:

```
rm -f test_fusion && make test_fusion CFLAGS="-D_GNU_SOURCE -O2 -Wall \
    -Wextra -std=c11 -pthread -Iinclude -DBENCH_SWEEP_ODR" && ./test_fusion
```

Accuracy is very nearly rate-independent. 3-D attitude RMS over the IMU axis, mag pinned at 100 Hz:

f_s (Hz)	12	26	52	104	208	416	**833**	1660	3332	6664	8000	16000	3
3-D att RMS	1.210°	1.163°	1.156°	1.181°	1.157°	1.176°	**1.178°**	1.192°	1.210°	1.224°	1.252°	1.280°	1
yaw att RMS	7.673°	2.370°	1.980°	2.036°	2.083°	2.105°	**2.185°**	2.203°	2.226°	2.241°	2.232°	2.241°	2
NIS_a (3-D)	15.66	7.40	3.84	2.11	1.26	0.82	**0.63**	0.53	0.47	0.42	0.42	0.35	0

±6% about the midpoint across a 2667× range in 3-D — 1.156° at 52 Hz to 1.299° at 32 kHz. A low-power board at 104 Hz gives up essentially nothing, which was the question that prompted the work.

The top two rungs arrived later, with the ODR-coverage work that added icm42688p's 16 kHz and 32 kHz. They did not change the conclusion, and they did not reveal a new mechanism either: 3-D attitude RMS has climbed steadily above the 833 Hz default for the whole length of this table, and it simply keeps climbing. Through 8 kHz the spread was ±4%; the two new rungs supply the rest. NIS_a keeps falling too, to 0.23 at 32 kHz — nearly 3× more under-confident than at the default, on the same trend the next paragraph describes rather than a break in it.

Worth separating from a related finding, because they are *not* the same effect. `test_heave_across_rates` measures float32 accumulation error in the heave integrator turning upward above 6664 Hz (0.112% there, 0.247% at 32 kHz; see the comment on that test). The MEKF's attitude degradation here begins at 833 Hz, well below that inflection, and is a property of the tuning across the ladder rather than of float32 accumulation. Two different curves that happen to both bend upward at the top.

But NIS_a is not flat, and crosses 1.0 near 150–200 Hz. The shipped measurement model is therefore most self-consistent around 200 Hz; at the 833 Hz default the filter is mildly *under*-confident, and below ~100 Hz increasingly over-confident. Accuracy does not track this at all.

And that is not a tuning choice. Re-running the (σ, τ) grid at 104 Hz: $\tau = 0.5$ s is still at the knee — the eight-fold drop in samples per correlation time, 416 to 52, does not move it — but *no point in the grid* brings NIS_a near 1. The best is $\approx$1.69 at $\sigma = 1.8$, twice the shipped value and at a real accuracy cost. The decisive evidence is `SCEN_GM`, the one configuration whose right answer is known in advance: with the knobs set to the truth it reports $\mathrm{NIS}_a = 0.87$ at 833 Hz and **$\approx$3.4 at 104 Hz**. Whatever this is, it is structural, and it is the open question this section leaves behind.

It is *not* the harness. The benchmark injects per-sample sigmas while the filter models a density × bandwidth, so the scenario scales its sensor draws by $\sqrt{f_s/833}$ (and the magnetometer independently by $\sqrt{f_{mag}/100}$). Cross-checked, R_a over-estimates the injected per-sample variance by 4.480× at 833 Hz and 4.480× at 104 Hz — identical to four figures.

The floor is at 12 Hz, only in yaw-only mode, and it is the gate.

f_s (Hz)	12	14	16	20	25	26	40	52
yaw att RMS	**7.673°**	3.949°	3.404°	2.838°	2.408°	2.370°	2.043°	1.980°
NEES(strict)	**402**	102	72.9	46.9	30.0	28.4	15.3	12.1
reject	**.1051**	.0000	.0000	.0000	.0000	.0000	.0000	.0000

Degradation from 52 Hz down to 14 Hz is smooth; 12 Hz is a cliff. The mechanism is visible in the last row — the gross-outlier gate goes from idle to rejecting 10.5% of updates, which is the rejection-feedback regime §4.7 describes: lost corrections cause drift, drift enlarges innovations, and those trip the gate more often still. 3-D mode at 12 Hz is unaffected (1.210°).

Note also that yaw-only is *best* around 40–52 Hz (1.980°), not at the 833 Hz default (2.185°).

This does not test the $\Phi = I + F_c\,\Delta t$ linearisation, and the claim in §4.4 that $\|\omega\|\Delta t \ll 1$ at supported ODRs remains unmeasured. The wave scenario's own peak rate is 18.8 °/s, so $\|\omega\|\Delta t = 0.027$ rad even at 12 Hz — nowhere near the limit. At the 2000 °/s full scale the config permits it would be 2.9 rad at 12 Hz. Probing that needs a high-rate-rotation scenario that does not exist.

Two further results worth having. The Gauss–Markov wave state is load-bearing at *every* rate, not a fix specific to 833 Hz: disabled, attitude RMS runs **5.0–11.4°** across the whole ladder — against 1.16–1.30° in 3-D with it enabled — and the gross-outlier gate goes from idle to rejecting **17–29%** of updates at 416 Hz and below. Both ranges are from the re-run that added the 16 kHz and 32 kHz rungs, and both are wider than the 8–11° and 12–27% previously recorded here; the mag-axis figures in the next paragraph reproduce to the digit against the same run, so this is a stale entry rather than a non-deterministic bench. Its *direction* also needs qualifying: the wave state mattering more as the rate falls is true of the rejection rate, not of accuracy. By RMS ratio it is worth least at 52 Hz (4.4×) and most around the 833 Hz default (8.6×), tapering to 7.8× at 32 kHz. And the magnetometer ladder is benign for accuracy — 3-D attitude is flat at 1.171–1.223° from a 1 Hz mag to a 1 kHz one — while being hostile to the NIS instrument: NIS_m runs 9.84 at 1 Hz against 0.46 at 1 kHz, because $R_m = N_m^2 f_{mag}$ shrinks with the rate while the attitude-error component of the innovation does not. NEES(strict) correspondingly *improves* at low mag rates (0.59 at 1 Hz against 5.74 at 100 Hz), which is independent corroboration of the dip-error diagnosis in §4.8.1: fewer 3-D mag updates inject less alignment dip bias into roll and pitch.

The gyro pad. `mekf_gyro_noise = 0.007` is a ~58× pad standing in for wave-induced angular dynamics. If that were all it covered it should be rate-invariant, since $Q_g = N_g^2\Delta t$ already delivers the same rad²/second at any rate. Measured, the RMS-optimal N_g is **0.002 at 833 Hz and 0.004 at 104 Hz** — eight times the Δt, twice the pad, close to the $\sqrt{\Delta t}$ that $N_g^2\Delta t$ implies for an error growing linearly in wall-clock time. That supports intra-sample rotation nonlinearity as (part of) what the pad absorbs.

It does **not** justify changing the default, for the reason §4.7 already documents for R_a: at 833 Hz, $N_g = 0.002$ gives the best attitude RMS in the grid (1.074° against 1.178° shipped) *and* the worst NEES(strict) near it (18.5 against 5.74). No scalar satisfies both. Worth recording that the shipped 0.007 is essentially optimal for yaw-only (2.185° against a best of 2.178°) while ~9% off optimal for 3-D — a deliberate compromise favouring the marine default, not an arbitrary number. Reproduce with `-DBENCH_SWEEP_NG`, composable with `-DBENCH_ODR_HZ=<rate>`.

4.8 Magnetometer update — `mekf_update_mag()` (`fusion.c:1438`)

Measurement in Gauss ($m = 0.01\, m_{\mu T}$). Predicted body field $h_{raw} = R^\top m_{ref}$, magnitude $|h_{raw}|$.

Normalization. Both predicted and measured vectors are unit-normalized before the update; the mag noise is rescaled $R_{m,n} = R_m/|h_{raw}|^2$ to preserve the optimal gain. (Rationale: in Gauss² units $\det S \approx R_m^3 \approx 4\times10^{-15}$ falls below the inverse's singularity threshold once P_{att} is small; normalizing makes S dimensionless and well-conditioned.) Residual $\nu = z - h$, res_sq = $\|\nu\|^2$.

Compass-health metrics — computed **before** the gates (§8).

Gates. Magnitude ratio $r = |m|/|h_{raw}|$ must lie in $[0.5, 2]$; direction res_sq < 4 ($\le$90° correction); and, once converged, the tight anomaly gate res_sq $>$ mag_reject_sq$/|h_{raw}|^2 \Rightarrow$ skip.

Reference adaptation (m_ref EMA), magnitude + dip only. When `mref_alpha`>0, `g_body_valid`, res_sq < 0.1, and $q_{quiet} < 2\times10^{-4}$ (platform calm), adapt the field magnitude and dip from **attitude-independent invariants**: $\sin(\text{dip}) = (m \cdot g_{body})/|m|$,

$$m^h_{ref} \leftarrow m^h_{ref}(1 + \alpha_{mref}(|m|\cos\text{dip}/m^h_{ref} - 1)), \quad m_{ref,z} \leftarrow m_{ref,z} + \alpha_{mref}(|m|\sin\text{dip} - m_{ref,z}).$$

The horizontal **direction** is never touched.

Update. If `mag_yaw_only` (marine default): rotate m to NED, form the heading innovation

$$y = \text{atan2}(m_y^{NED}, m_x^{NED}) - \text{atan2}(m_{ref,y}, m_{ref,x}) \text{ (wrapped)},$$

noise $R_\psi = R_m/(m^h_{ref})^2$, and call the scalar update §4.6. Otherwise call the full vector update §4.5 with $(h, z, R, 11.34)$, where R carries the anisotropic dip term of §4.8.1. A near-vertical field ($m^h < 0.2|h_{raw}|$) carries no heading information and is skipped.

4.8.1 The dip-reference error, and the anisotropic R

In 3-D vector mode the field's **dip** constrains roll and pitch. That is the mode's whole value over `mag_yaw_only`, and also its exposure: the dip of m_{ref} is the one part of the reference the filter cannot establish accurately on its own.

m_{ref} is fixed once, at alignment, from the tilt estimate available at that moment (§4.3). In a seaway that tilt is wrong, and the error is baked in permanently — in 3-D mode it becomes a **constant roll/pitch bias** that P has no term for, which is what a large NEES(strict) is reporting. Measured over the 12-seed benchmark — these are the authoritative 3-D figures:

3-D mode	dip error	att RMS	hdg RMS	NEES(tr)	NEES(st)
align from 1 sample	−4.38°	4.452°	0.828°	3.47	335
align over the window	+0.86°	1.204°	0.745°	0.21	12.8
+ dip reference exact (WMM)	0.00°	0.841°	0.737°	0.11	0.22

The per-axis decomposition is unambiguous: at the top row the mean attitude error is $[-3.85°, -2.00°, -0.28°]$ against an RMS of $[3.87°, 2.04°, 0.87°]$ — almost pure bias, not spread.

It cannot be healed in run. The m_ref EMA above exists for exactly this and is gated on $q_{quiet} < 2\times10^{-4}$, which a seaway never reaches ($q_{quiet} \approx 5\times10^{-3}$). That gate is correct, not an oversight. Raising it looks like a fix over 120 s and is refuted over 30 minutes:

3-D, 1800 s window	dip error	$\lvert m^h_{ref}\rvert$ error	att RMS	NEES(st)
gate 2×10^{-4} (shipped)	$+0.862°$	-5.04%	1.151°	12.84
gate 2×10^{-2}	$-1.460°$	$+4.24\%$	1.725°	42.10
gate off	$-1.469°$	$+4.27\%$	1.729°	42.31

The reference sails past truth and keeps going, because the samples clearing the $|a|$ band are wave-phase correlated: learning from that subset walks m_{ref} away rather than toward. (De-contaminating g_{body} with the Gauss–Markov $\hat{a}_w$ was also tried, and is worse — $\hat{a}_w$ has itself absorbed some tilt, so adding it back re-injects a correlated error.)

So the dip error is **not observable from seaway data**. It is removed at the source by WMM invariants (`mekf_set_mref_invariants`), or it is admitted into P.

The anisotropic term. A dip error $d\delta$ rotates m_{ref} about $\hat{a}_{NED} = \hat{h}_{hor} \times \hat{e}_D$, so in body frame it perturbs the normalised prediction along one unit tangent direction:

$$\hat{a}_{body} = R^\top \hat{a}_{NED}, \qquad u = \hat{a}_{body} \times \hat{h}, \qquad \|u\| = 1, \quad u^\mathsf{T}\hat{h} = 0,$$

(the last two because $\hat{m}_{ref}$ lies in the plane spanned by $\hat{h}_{hor}$ and $\hat{e}_D$, hence $\hat{m}_{ref} \perp \hat{a}_{NED}$). The uncertainty is therefore exactly rank-1:

$$R = R_{m,n} I_3 + \sigma_{dip}^2\, uu^\mathsf{T},$$

with σ_{dip} = `mekf_mag_dip_sigma_deg` in radians (a dip error of δ displaces the predicted unit direction by δ, so the units match directly). Default 1.0°, which is the measured +0.86° residual rounded up — **not** a value fitted to the metric.

Why rank-1 and not just a larger R_m. Isotropic inflation reaches the same NEES: $R_m \times 64$ gives NEES(strict) 12.83 → 1.57 with attitude and heading both marginally better. But it deweights the heading-carrying components too, and `nis_mag` falls to **0.01** — the wire's magnetometer-health instrument stops being able to report a fault. The rank-1 form leaves those components alone. Sweep, 3-D mode with a window-aligned reference:

σ_{dip}	att RMS	hdg RMS	NEES(st)	`nis_mag`
0°	1.204°	0.745°	12.83	0.52
0.5°	1.186°	0.723°	9.06	0.39
1.0° (shipped)	**1.178°**	**0.714°**	**5.74**	**0.31**
2.0°	1.177°	0.715°	3.14	0.28
3.0°	1.179°	0.720°	2.15	0.27

dof = 2 survives. Because u is tangent, $R\hat{h} = R_{m,n}\hat{h}$, so $\hat{h}$ remains an eigenvector of $S = HPH^\top + R$ with eigenvalue $R_{m,n}$; combined with $H^\top\hat{h} = 0$ the §8.2 derivation $E[d^2] = 3 - R_{m,n}\hat{h}^\top S^{-1}\hat{h} = 2$ is unchanged. An anisotropy with any radial component would break it silently — and would also stop the term working at all, since the radial direction of a normalised measurement carries no information (measured: a contaminated u widens P by 1% instead of 9.5% and absorbs 53% of the pull instead of 93%).

Note `nis_mag` settling near **0.5** once σ_{dip} dominates is the correct reading, not a regression: one of the two tangent degrees of freedom is deliberately deweighted, so a consistent filter reads about half.

What this does not do. It does not reach NEES(strict) = 1, and cannot. The dip error is a *bias*; a covariance term can only partly stand in for one, and the accelerometer legitimately keeps P tight in roll and pitch. Driving the number to 1 would need $\sigma_{dip} \approx 4°$, i.e. inventing uncertainty to satisfy a statistic. For an install that cares, the answer is a position source.

As-implemented / audit note. Adapting **only** magnitude and dip is deliberate: the horizontal direction *is* the heading reference, so learning it from the filter's own attitude would feed the heading estimate back into its own datum and let both random-walk together (gauge drift). The adaptation targets are body-frame invariants (total magnitude; dip from the mag–gravity angle), so the filter's attitude never enters. Heading-only fusion is the default because the soft-iron matrix is fitted from a 2-D swing (§3.3, §12.3) and its dip channel is least trustworthy.

Source: heading-only / tilt-compensated magnetometer fusion — standard marine AHRS practice, Farrell 2008 *(canonical)*; gauge-feedback avoidance is imud-specific design (see the in-code rationale in `mekf_update_mag`, `fusion.c:1550`).

4.9 WMM reference invariants — `mekf_set_mref_invariants()` (`fusion.c:1057`)

Given WMM horizontal magnitude H and vertical Z (Gauss) at a known position (from §13), rescale `m_ref` to those invariants while preserving horizontal direction: with $m_h = \sqrt{m_{ref,x}^2 + m_{ref,y}^2}$ and $s = H/m_h$,

$$m_{ref,x} \leftarrow s\, m_{ref,x}, \quad m_{ref,y} \leftarrow s\, m_{ref,y}, \quad m_{ref,z} \leftarrow Z.$$

No-op before alignment or if $H \le 0$.

Source: imud-specific (WMM supplies the invariants of §13).

4.10 State extraction — `mekf_get_state()` (`fusion.c:1988`)

Euler angles from $R(q)$ (NED 3-2-1 aerospace):

$$\theta = \arcsin(-R[2][0]), \quad \phi = \operatorname{atan2}(R[2][1], R[2][2]), \quad \psi = \operatorname{atan2}(R[1][0], R[0][0]).$$

Magnetic heading ψ wrapped to $[0, 360°)$. Attitude covariance (`cov[9]`) $= P[0{:}3, 0{:}3]$; bias variance $=$ $\operatorname{diag} P[3{:}6]$; `quiescence` $= q_{quiet}$. `rate_of_turn` is left 0 here and filled by the fusion thread (§5).

Source: quaternion→Euler, Diebel 2006 *(canonical)*.

4.11 Reconfigure — `mekf_reconfigure()` (`fusion.c:1918`)

Recomputes Q_g, Q_b, R_a, R_m, the skip band, `mag_reject_sq`, `mag_yaw_only`, `mref_alpha`, `conv_thresh`, and the gate-health EMA rate from a new config on hot-reload; q, P, b, dt and the runtime scalars (`Ra_scale`, `acc_quiet_ema`, the health EMAs) are untouched, so the filter keeps running. Both this and `mekf_init()` derive their tuning through one shared helper (`mekf_derive_tuning`, `fusion.c:864`), so no value can be updated in one path and forgotten in the other — `mref_alpha` previously was, leaving the m_ref EMA at a stale rate after any `mag_odr_hz` change.

---

5. Euler rates and rate of turn

Computed in the fusion thread (`fusion_thread`, `imu.c:1429–1476`) from the bias-corrected body rate $\omega = s.\text{gyro} - b$ and the current Euler angles, using the inverse of the 3-2-1 kinematic relation.

Roll rate (for sea-state, §7), with a near-gimbal-lock fallback:

$$\dot\phi = \omega_x + \tan\theta\,(\omega_y \sin\phi + \omega_z \cos\phi), \qquad \dot\phi = \omega_x \text{ if } |\cos\theta| \le 0.2.$$

Pitch rate (no singularity):

$$\dot\theta = \omega_y \cos\phi - \omega_z \sin\phi.$$

Rate of turn (heading rate, output in $\deg \cdot \min^{-1}$):

$$\dot{\psi} = \frac{\omega_y \sin\phi + \omega_z \cos\phi}{\cos\theta}, \qquad \dot{\psi} = \omega_z \text{ if } |\cos\theta| \le 0.2, \qquad \mathrm{ROT} = \dot{\psi} \cdot \frac{180}{\pi} \cdot 60.$$

As-implemented. ROT is the *Euler yaw rate*, not raw body-Z gyro: at 20° roll the body-Z rate under-reads heading rate and couples in pitch rate, which autopilots notice. Near ±90° pitch ($|\cos\theta| \le 0.2$) the code falls back to the body-axis rate to avoid division blow-up.

Source: 3-2-1 Euler kinematics, Titterton & Weston 2004 §3.6; Diebel 2006 *(canonical)*.

---

6. Heave estimator — `heave_update()` (`fusion.c:1771`)

Vertical displacement from vertical acceleration, by leaky double integration followed by a true first-order high-pass.

Vertical acceleration (NED down, zero at rest):

$$a_D = \big(R(q)\, a_b\big)_D + g = R[2][:] \cdot a_b + g.$$

Leaky double integration, leak $\ell = dt/\tau$:

$$v \leftarrow (v + a_D\, dt)(1-\ell), \qquad d \leftarrow (d + v\, dt)(1-\ell).$$

Output high-pass (exact zero at DC), $\alpha_{hp} = \tau/(\tau + dt)$:

$$y \leftarrow \alpha_{hp}\,(y + d - d_{prev}), \qquad d_{prev} \leftarrow d,$$

heave (positive **up**) $= -y$. `heave_rate` output $= -v$. Settled after $\ge 10\tau$ (`settled` flag).

As-implemented / audit note. Two leaky integrators bound drift but retain a DC gain of $\sim \tau^2$ (≈ 144 m per $\mathrm{m \cdot s^{-2}}$ at the default $\tau = 12$ s); any slow residual (accel bias, small tilt error, start-up transient) would appear at metre scale. The output high-pass has a **structural** zero at DC, killing those residuals rather than estimating them. Passband amplitude error $\approx 2\%$ at 8 s, $\approx 4\%$ at 12 s for $\tau = 12$ s (band-limited: very long-period swell is attenuated). $\tau = 0$ disables. The estimate is a filtered kinematic quantity, not a statistically-optimal one.

Source: leaky integrator + first-order high-pass are standard discrete-time filter forms, Oppenheim & Schafer 2010; the double-integration- plus-high-pass heave technique is long-standing in wave-buoy practice (Datawell / Longuet-Higgins tradition) *(canonical)*.

---

7. Sea-state statistics — `seastate_*` (`fusion.c:1829–1916`)

Windowed spectral moments over the heave, roll, and pitch oscillations via exponentially-weighted mean/variance pairs — no FFT, no sample storage.

EW mean/variance recursion (`ew_stat`, `fusion.c:1847`), $\alpha = dt/\tau$:

$$\mu \leftarrow \mu + \alpha\, d, \qquad \sigma^2 \leftarrow \sigma^2 + \alpha\big((1-\alpha)\, d^2 - \sigma^2\big), \qquad d = x - \mu.$$

Applied to six signals: heave, heave-rate, roll, roll-rate, pitch, pitch-rate. Fed **only while heave is settled** (`seastate_update`, `fusion.c:1854`).

Outputs (from the variances; $m_0 = \mathrm{var}(x)$, $m_2 = \mathrm{var}(\dot{x})$):

$$H_s = 4\sqrt{m_0^{heave}}, \qquad T_z = 2\pi\sqrt{\frac{m_0^{heave}}{m_2^{heave}}}, \qquad T_{roll} = 2\pi\sqrt{\frac{m_0^{roll}}{m_2^{roll}}},$$

$$A_{roll} = 2\sqrt{m_0^{roll}}, \qquad A_{pitch} = 2\sqrt{m_0^{pitch}}, \qquad T_{pitch} = 2\pi\sqrt{\frac{m_0^{pitch}}{m_2^{pitch}}}.$$

Guards: all return 0 until settled ($\geq 2\tau$); periods additionally return 0 below the oscillation floors σ(heave) < 2 cm (`SEASTATE_MIN_HEAVE_SIG`) or σ(angle) < 0.3° (`SEASTATE_MIN_ANGLE_SIG`), or non-positive rate variance.

As-implemented / audit note. $H_s = 4\sqrt{m_0}$ is the significant wave height for a narrow-band Gaussian sea (Longuet-Higgins). The period identity $T = 2\pi\sqrt{m_0/m_2}$ is Rice's mean up-crossing period from the zeroth and second spectral moments — **exact for a pure sine** (var $= A^2/2$, var($\dot{x}$) $= A^2\omega^2/2$, ratio $= 1/\omega^2$) and an estimator for a real seaway. Amplitudes are reported as **significant single amplitudes** (2σ, seakeeping convention); wave height is the significant **double** amplitude (4σ). The EW variance is biased toward the window's recent history and toward the EW mean (so a steady heel / trim is removed from roll/pitch, appearing in μ, not σ). The default $\tau = 120$ s is far shorter than the 10–20 min oceanographic record; it favours a responsive live display. Roll/pitch **rates** are Euler rates (§5), not body rates.

Source: $H_s = 4\sqrt{m_0}$ Longuet-Higgins 1952; spectral-moment up-crossing period Rice 1944–45; engineering treatment Tucker & Pitt 2001; roll-period/seakeeping convention Lloyd 1989 *(canonical)*. EW variance recursion, West 1979 / Finch 2009 *(canonical)*.

---

8. Compass-health diagnostics — `mekf_update_mag()` (`fusion.c:1438`)

Two EMAs ($\alpha = 1/3000$, $\tau \approx 30$ s at 100 Hz mag ODR), updated **before** the rejection gates so that gated-out anomalies still register:

Field-magnitude anomaly (attitude-independent): $$ \text{mag_anom} \leftarrow \text{mag_anom} + \alpha\left(\frac{\lvert, |m|-|h_{raw}|,\rvert}{|h_{raw}|}

- \text{mag_anom}\right). $$

Heading residual (mode-independent; $m^{NED} = R(q)\,m$):

$$y = \text{atan2}(m_y^{NED}, m_x^{NED}) - \text{atan2}(m_{ref,y}, m_{ref,x}) \text{ (wrapped to } \pm\pi), \qquad \text{mag\_resid} \leftarrow \text{mag\_resid} + \alpha(|y| - \text{mag\_resid}),$$

gated on both horizontal magnitudes exceeding $0.2|h_{raw}|$.

As-implemented. Diagnostics only — they never modify the filter. Fed from pre-gate innovations because the rejected samples are precisely the anomalies these metrics exist to surface. The 30 s time constant scales inversely with a non-standard mag rate (acceptable for a health indicator).

8.1 Update-gate health — `gate_health()` (`fusion.c:159`)

Two further EMAs ($\tau \approx 30$ s, gain per §4.7), written by every `eskf_update` / `eskf_update_yaw` call — accepted, capped and rejected alike, with $\gamma_r = 25\gamma$ the gross-reject threshold:

$$\text{innov\_weight} \leftarrow \text{innov\_weight} + \alpha(w - \text{innov\_weight}), \qquad w = \begin{cases} 1 & d^2 \leq \gamma \\ \sqrt{\gamma/d^2} & \gamma < d^2 \leq \gamma_r \\ 0 & d^2 > \gamma_r \end{cases}$$

$$\text{innov\_reject} \leftarrow \text{innov\_reject} + \alpha(\mathbb{1}[d^2 > \gamma_r] - \text{innov\_reject}).$$

`innov_weight` near 1 means the filter is accepting its measurements at face value; a sustained drift toward 1/5 means the Huber cap is doing continuous work, which is the regime in which the covariance inconsistency noted in §4.5 matters. `innov_reject` separates "capping hard" from "throwing measurements away outright".

Both are exported on the wire (§8 of `spec.md`, offsets 256 and 260) so the behaviour can be observed in the field rather than inferred from a synthetic benchmark.

Source: exponential moving average, standard *(canonical)*; innovation consistency monitoring Bar-Shalom et al. 2001 *(canonical)*.

8.2 Measurement-model consistency — `nis_record()` (`fusion.c:197`)

Where §8.1 reports how hard the robustness machinery is *working*, this reports whether the noise model underneath it is *right*. Two EMAs of the normalised innovation squared, per channel (wire offsets 264 and 268):

$$ \text{nis} \leftarrow \text{nis} + \alpha\Big(\min\big(d^2/n_{dof}„100\big)

- \text{nis}\Big),\qquad d^2=\nu^\top S^{-1}\nu. $$

Accumulated **before** the Huber cap and **including** rejected updates: the cap censors d^2 at γ, so a post-cap average would be bounded by construction and could never report the inconsistency it exists to measure. The channels are split because they run at very different rates (833 Hz accel vs ~104 Hz mag); a combined EMA would be ~8:1 accel.

Degrees of freedom. The vector updates carry $n_{dof} = 2$, not 3, because the measurement is a *unit* vector — normalising z removes the radial component, so the noise lives only in the tangent plane. Write $\hat{z}$ for the predicted direction ($\hat{z} = h$ without the wave state, $\hat{z} = (h - \hat{a}_w)/\|h - \hat{a}_w\|$ with it, §4.5.1). The true innovation covariance is $E[\nu\nu^\top] = HPH^\top + R(I - \hat{z}\hat{z}^\top)$ while $S = HPH^\top + R\,I$:

$$E[d^2] = \mathrm{tr}(S^{-1}E[\nu\nu^\top]) = \mathrm{tr}(S^{-1}(S - R\,\hat{z}\hat{z}^\top)) = 3 - R\,\hat{z}^\top S^{-1}\hat{z},$$

and in **both** cases $H^\top\hat{z} = 0$ — for the plain form because $H = [h]_\times$ gives $[h]_\times h = 0$, and for the wave-aware form because every row of H is left-multiplied by the tangent projector $P_{\hat{z}} = I - \hat{z}\hat{z}^\top$ and $P_{\hat{z}}\hat{z} = 0$. So $\hat{z}$ is an eigenvector of S with eigenvalue R, $S^{-1}\hat{z} = \hat{z}/R$, and the correction term is exactly 1:

$$E[d^2] = 2 \quad \text{for both the gravity and 3-D mag updates.}$$

The yaw-only scalar update has $n_{dof} = 1$. Dividing by 3 instead would peg a perfectly consistent filter at 0.67.

This identity is the reason the projector in §4.5.1 cannot be dropped: it is the *only* thing that keeps $H^\top\hat{z} = 0$ once $\hat{a}_w \neq 0$, and without it the wire field silently stops meaning what its documentation says.

Measured values over the wave benchmark are 1.01 (3-D) / 0.69 (yaw-only) with the shipped Gauss–Markov wave state — i.e. the model is now consistent, and mildly conservative in yaw-only mode. They were 19 / 25 before §4.7.1, and 56 / 57 with the `m33_inv` fix but no wave state. `imud-cal fit-ra` computes the same statistic offline from a capture, through the filter's own `mekf_accel_probe()`, so the live number can be checked against real water.

Source: NIS consistency test Bar-Shalom et al. 2001 §5.4 *(canonical)*.

---

9. Engine-vibration detector — `ism_reader_thread()` (`imu.c`)

EMA of squared specific-force deviation from 1 g, stepped once per sample:

$$e \leftarrow e + \alpha((\|a\| - g)^2 - e), \qquad \alpha = \frac{1}{\tau\, f_{ODR}}, \quad \tau = 1\,\text{s}.$$

α is derived from the **resolved sample rate**, not fixed. Because the EMA advances once per sample but samples arrive in FIFO bursts, a constant α made the effective time constant a function of burst depth (a deeper burst = more steps per read = a faster EMA) rather than the intended 1 s.

Assertion uses **hysteresis** rather than a bare threshold:

$$\text{engine\_on} = \begin{cases} \text{true} & e > \text{engine\_vibration\_g2} \\ \text{false} & e < 0.7\,\text{engine\_vibration\_g2} \\ \text{unchanged} & \text{otherwise} \end{cases}$$

A single threshold chattered at the boundary, and each toggle steps `Ra_scale`, rewrites the skip band, emits a log line, and flips `FLAG_ENGINE_ON` on the wire — all externally visible. `engine_on` raises the accel noise $\times 4$ and widens the skip band (§4.7); the fusion thread re-asserts both on every sample, so a config reload cannot leave the skip band and `Ra_scale` disagreeing.

Source: EMA threshold detector with Schmitt hysteresis, standard *(canonical)*.

---

10. Startup gyro-bias estimation — `fusion_thread()` (`imu.c:934`)

Mean of the gyro over a still window (N = `gyro_bias_sec` $\cdot$ ODR samples):

$$\hat{b}_k = \frac{1}{N} \sum_{i=1}^{N} \omega_{i,k}.$$

A per-axis motion check computes the sample standard deviation $\sigma_k = \sqrt{\overline{\omega^2} - \bar{\omega}^2}$; if $\max_k \sigma_k > 0.00873$ $\text{rad} \cdot \text{s}^{-1}$ ($0.5° \cdot \text{s}^{-1}$) the window is **doubled once** (a longer average spans more wave cycles). The result seeds `mekf_init` (§4.2); the MEKF refines it online regardless.

Source: sample mean/variance, elementary *(canonical)*.

---

11. Timestamp anchoring and per-sample dt — `imu_math.c:58–194`

A single anchor (chip_ticks, wall_ns, tai_ns, gen) maps the sensor's free-running counter to system time. Per sample (`chip_to_wall`, `imu_math.c:178`), with `tick_ns` the counter period:

$$\text{offset} = (\text{chip\_ts} - \text{chip\_ticks}) \cdot \text{tick\_ns}, \qquad t_{wall} = \text{wall\_ns} + \text{offset},$$

t_{tai} likewise. Subtraction is unsigned 32-bit (wrap-safe within one anchor interval). The anchor is refreshed on first read, every 60 s with a hardware counter, or every burst without one.

Per-sample interval for prediction (`imu.c`): the wall-time delta between consecutive samples of the same anchor generation, **clamped to** $[0.5, 2]\times$ **nominal**; outside that (FIFO gap, anchor reset) the nominal 1/ODR is used. This keeps oscillator tolerance (few % on the chip clock) from scaling integrated rotation while rejecting gap artefacts.

As-implemented. Linear (constant-rate) interpolation between anchors; no clock-drift model beyond the 60 s re-anchor. `tick_ns`=0 for chips without a hardware timestamp degenerates the offset to 0 (the anchor wall-time is then the per-sample time).

Source: imud-specific timestamp reconstruction.

---

12. Offline calibration fits (`imud-cal`, `cal_math.c`)

12.1 Linear solver — `gauss4()` (`cal_math.c:25`)

Gaussian elimination with **partial pivoting** on the augmented 4×5 system, then back-substitution. Returns -1 if any pivot $< 10^{-12}$. Double precision.

Source: Golub & Van Loan 2013 §3.4 *(canonical)*.

12.2 Hard-iron sphere fit — `sphere_fit()` (`cal_math.c:153`)

Algebraic (linearized) sphere fit. The sphere $(x-c_x)^2+(y-c_y)^2+(z-c_z)^2=r^2$ is linearized as

$$x^2+y^2+z^2 = 2c_x x + 2c_y y + 2c_z z + (r^2-\|c\|^2),$$

and the normal equations $A^\top A\,p = A^\top b$ are accumulated incrementally (`sphere_add`, `cal_math.c:59`) over sums $\Sigma x, \Sigma x^2, \Sigma xy, \Sigma xr^2, \ldots$. Solved by `gauss4`:

$$A^\top A = \begin{bmatrix} 4S_{xx} & 4S_{xy} & 4S_{xz} & 2S_x \\ 4S_{xy} & 4S_{yy} & 4S_{yz} & 2S_y \\ 4S_{xz} & 4S_{yz} & 4S_{zz} & 2S_z \\ 2S_x & 2S_y & 2S_z & n \end{bmatrix}, \quad A^\top b = \begin{bmatrix} 2S_{xr} \\ 2S_{yr} \\ 2S_{zr} \\ S_{xx}+S_{yy}+S_{zz} \end{bmatrix},$$

center $c = p[0{:}3]$, $r=\sqrt{p_3+\|c\|^2}$. Center = hard-iron offset; r = reference field magnitude for §12.3.

As-implemented. This is the *algebraic* fit (minimizes an algebraic residual, not orthogonal geometric distance); fast, closed-form, and adequate for well-sampled swings, but statistically biased for sparse/noisy data versus a geometric fit.

Source: algebraic sphere/magnetometer fit, Renaudin et al. 2010; Coope 1993 *(canonical).*

12.3 Soft-iron 2-D ellipse fit — `ellipse_fit()` (`cal_math.c:106`)

Conic least-squares on the horizontal magnetometer locus $Ax^2+Bxy+Cy^2=1$ (regressors x^2, xy, y^2; normal equations from $S_{40}, S_{31}, S_{22}, S_{13}, S_{04}, S_{20}, S_{11}, S_{02}$ via `ellipse_add`, solved by `gauss4` with an identity pad row). Requires the conic $M = \begin{pmatrix} A & B/2 \\ B/2 & C \end{pmatrix}$ positive-definite ($A>0$, $\det>0$).

Soft-iron matrix $S = r\cdot M^{1/2}$ via the 2×2 symmetric eigendecomposition: eigenvalues $\lambda_{1,2} = \frac{A+C}{2} \pm \sqrt{\frac{1}{4}(A-C)^2+\frac{1}{4}B^2}$, unit eigenvector v for λ_1, and

$$S = s_1\,vv^\top + s_2\,v_\perp v_\perp^\top, \qquad s_{1,2} = r\sqrt{\lambda_{1,2}},\ v_\perp = (-v_y, v_x).$$

As-implemented / audit note. The soft-iron correction is **2-D** (X–Y plane only). This is a deliberate specialization for the heading-only fusion default (§4.8): the swing is a horizontal circle, so the vertical/dip channel is not calibrated. `ellipse_fit` includes a diagonal fallback when the conic is degenerate. The full `mag_soft_iron[3][3]` applied live (§3.3) therefore has its Z couplings from configuration/identity, not from this fit.

Source: algebraic conic/ellipse fit, Fitzgibbon, Pilu & Fisher 1999; magnetometer soft-iron, Renaudin et al. 2010 *(canonical).*

12.4 Heading-circle coverage — `cal_cov_mark()` (`cal_math.c:308`)

Guided-swing coverage: the heading circle is split into `nsec` sectors; the sample (x,y) relative to center (c_x, c_y) is binned by

$$s = \left\lfloor \frac{\operatorname{atan2}(y-c_y,\,x-c_x) \bmod 2\pi}{2\pi/\mathrm{nsec}} \right\rfloor.$$

`cal_cov_count` sums the marked sectors. Pure bookkeeping (no fit).

12.5 Allan variance — `allan_deviation()` (`cal_math.c:327`)

Overlapping Allan deviation. With the cumulative integral $\theta[k] = \sum_{j<k} x_j\,dt$ and octave cluster lengths $m = 1, 2, 4, \ldots$ ($\tau = m\,dt$):

$$\sigma^2(\tau) = \frac{1}{2\tau^2\,(N-2m+1)} \sum_{k=0}^{N-2m} \big(\theta[k{+}2m] - 2\theta[k{+}m] + \theta[k]\big)^2.$$

Characterization (`allan_characterize`, `cal_math.c:360`): white-noise density from the shortest cluster,

$$N = \sigma(\tau_0)\sqrt{\tau_0},$$

bias instability from the curve minimum,

$$B = 0.664 \cdot \min_{\tau} \sigma(\tau).$$

As-implemented. Octave (not fully-overlapping all-τ) spacing; N is read at τ_0 (assumes white noise dominates there); B uses the IEEE-952 factor 0.664 at the deviation minimum (an upper bound if the flat region is not reached). These characterize the sensor; they do **not** feed the filter tuning (§4.2 note).

Source: Allan 1966; overlapping estimator Riley 2008 (NIST SP 1065); N/B extraction and 0.664, IEEE Std 952 *(code comment: "IEEE 952")*.

12.6 Gyro-bias / temperature fit — `gyro_temp_fit()` (`cal_math.c:384`)

Ordinary least-squares line of gyro bias vs. temperature about a reference T_{ref}: with $t_i = T_i - T_{ref}$,

$$\text{coeff} = \frac{n\sum t_i y_i - \sum t_i \sum y_i}{n\sum t_i^2 - (\sum t_i)^2}, \qquad \text{bias\_ref} = \frac{\sum y_i - \text{coeff}\sum t_i}{n}.$$

Rejects fits with temperature span < 1 °C. `coeff` → `gyro_temp_coeff`, applied live in §3.2.

Source: ordinary least squares, elementary *(canonical)*.

---

13. World Magnetic Model — `wmm_field_ned()` (`wmm.c:119`)

Degree/order 12 spherical-harmonic geomagnetic field. This section states the implemented computation; the defining theory is the WMM report cited below.

Secular variation to the decimal year t (`wmm_decimal_year`, `wmm.c:73`), $\Delta t = t - t_{epoch}$:

$$g_n^m(t) = g_n^m + \dot{g}_n^m\,\Delta t, \qquad h_n^m(t) = h_n^m + \dot{h}_n^m\,\Delta t.$$

Geodetic (WGS-84) → geocentric spherical. With prime-vertical radius $N = a/\sqrt{1 - e^2\sin^2\varphi}$ ($a =$ 6378137 m, e^2 from $f = 1/298.257223563$):

$$p_x = (N + \text{alt})\cos\varphi, \quad p_z = (N(1-e^2) + \text{alt})\sin\varphi,$$

$$r = \sqrt{p_x^2 + p_z^2}, \quad \theta = \arccos(p_z/r) \text{ (co-latitude)}, \quad \lambda = \text{lon}.$$

Schmidt quasi-normalized associated Legendre functions P_n^m and derivatives $dP_n^m/d\theta$, as coded (`wmm.c:178`): seeds $P_0^0 = 1$, $P_1^0 = \cos\theta$, $P_1^1 = \sin\theta$; for $n \geq 2$,

$$P_n^n = \sqrt{\tfrac{2n-1}{2n}}\,\sin\theta\,P_{n-1}^{n-1}, \qquad P_n^m = \frac{2n-1}{\sqrt{n^2-m^2}}\cos\theta\,P_{n-1}^m - \sqrt{\tfrac{(n-m-1)(n+m-1)}{(n-m)(n+m)}}\,P_{n-2}^m,$$

the second covering zonal ($m = 0$), sub-diagonal ($m = n-1$, second term 0), and tesseral terms; derivatives follow by differentiating the same recursion.

Field synthesis (geocentric NED), $a = 6371.2$ **km:**

$$X_{gc} = \sum_{n=1}^{12}\left(\tfrac{a}{r}\right)^{n+2}\sum_{m=0}^{n}\left(g_n^m\cos m\lambda + h_n^m\sin m\lambda\right)\frac{dP_n^m}{d\theta},$$

$$Y_{gc} = \frac{1}{\sin\theta} \sum_{n=1}^{12} \left(\frac{a}{r}\right)^{n+2} \sum_{m=0}^{n} m(g_n^m \sin m\lambda - h_n^m \cos m\lambda)\, P_n^m,$$

$$Z_{gc} = -\sum_{n=1}^{12} (n+1) \left(\frac{a}{r}\right)^{n+2} \sum_{m=0}^{n} (g_n^m \cos m\lambda + h_n^m \sin m\lambda)\, P_n^m.$$

Geocentric → geodetic NED rotation by $\delta = \varphi - \varphi_{gc}$ ($\varphi_{gc} = \pi/2 - \theta$):

$$X_{NED} = X_{gc}\cos\delta + Z_{gc}\sin\delta, \quad Y_{NED} = Y_{gc}, \quad Z_{NED} = Z_{gc}\cos\delta - X_{gc}\sin\delta.$$

Declination (`wmm_declination`, `wmm.c:251`): $D = \operatorname{atan2}(Y_{NED}, X_{NED})$. The horizontal magnitude $H = \sqrt{X^2 + Y^2}$ and vertical Z supply the §4.9 magnetic invariants.

As-implemented. Secular variation is linear extrapolation from the epoch coefficients (valid over the model's 5-year validity window). `wmm_load` requires exactly the 90 degree-12 coefficient rows and rejects truncated files. Powers $(a/r)^{n+2}$ are formed by iterative multiplication, not `pow`. Poles are guarded by clamping $\sin\theta \geq 10^{-10}$.

Source: WMM2025 Technical Report, Chulliat et al. 2024 (NCEI/BGS) — the defining document; ALF recursion Langel 1987 / WMM Technical Note 28 *(code comment)*; WGS-84 NIMA TR8350.2 *(canonical)*.

---

14. Symbol ↔ code map

Symbol	Meaning	Code
q	attitude quaternion, body→NED, scalar-first	`mekf_t.q[4]`
b	gyro bias	`mekf_t.bias[3]`
a_w	wave acceleration, body, g units	`mekf_t.wave_acc[3]`
P	9×9 error covariance $[\delta\theta \mid \delta b \mid \delta a_w]$	`mekf_t.P[MEKF_N][MEKF_N]`
$R(q)$	body→NED rotation matrix	`q_to_R()`
Q_g, Q_b	process-noise variances / step	`mekf_t.Qg, .Qb`
R_a, R_m	accel / mag meas. noise var	`mekf_t.Ra, .Rm`
σ, τ	Gauss–Markov wave σ (m · s^{-2}), correlation time (s)	`mekf_wave_accel`, `mekf_wave_accel_tau_s`
σ_g^2, φ, Q_w	wave variance (g^2), GM transition, GM noise	`mekf_t.wave_sig2`, `mekf_predict`
$P_{\hat{z}}$	tangent projector $I - \hat{z}\hat{z}^\top$	`wave_jacobian`
H	measurement Jacobian	`eskf_update`, `eskf_update_yaw`
γ	χ^2 innovation gate	`ACCEL_CHI2_GATE`=11.34, `YAW_CHI2_GATE`=6.63
m_{ref}	Earth-field reference, NED, Gauss	`mekf_t.m_ref[3]`
q_{quiet}	quiescence EMA $(	a
g_{body}	accel gravity direction, body	`mekf_t.g_body[3]`
H_s, T_z	sig. wave height, zero-cross period	`seastate_wave_height/_period`
a_D	NED-down linear accel	`heave_update`
N, B	Allan noise density, bias instability	`allan_characterize`
g_n^m, h_n^m	Gauss coefficients	`wmm_t.g/.h`, `.g_sv/.h_sv`
D	magnetic declination	`wmm_declination`

---

15. References

Citations are to the canonical source for each method; imud implements standard techniques rather than novel mathematics. Items the maintainer may wish to verify against a specific edition/page are flagged [**verify**].

1. **J. Solà**, "Quaternion kinematics for the error-state Kalman filter," arXiv:1711.02508, 2017. *(MEKF, quaternion kinematics, error-state covariance — named in `fusion.c`.)*
2. **F. L. Markley, J. L. Crassidis**, *Fundamentals of Spacecraft Attitude Determination and Control*, Springer, 2014. *(MEKF.)*
3. **N. Trawny, S. Roumeliotis**, "Indirect Kalman Filter for 3D Attitude Estimation," Univ. Minnesota TR-2005-002. *(error-state attitude EKF.)*
4. **R. E. Kalman**, "A New Approach to Linear Filtering and Prediction Problems," *J. Basic Eng.*, 1960. *(Kalman update.)*
5. **R. S. Bucy, P. D. Joseph**, *Filtering for Stochastic Processes with Applications to Guidance*, 1968. *(Joseph-form covariance.)* **[verify]**
6. **P. J. Huber**, "Robust Estimation of a Location Parameter," *Ann. Math. Statist.*, 1964. *(innovation influence capping.)*
7. **Y. Bar-Shalom, X.-R. Li, T. Kirubarajan**, *Estimation with Applications to Tracking and Navigation*, Wiley, 2001. *(Mahalanobis χ^2 gating.)*
8. **J. A. Farrell**, *Aided Navigation: GPS with High Rate Sensors*, McGraw-Hill, 2008. *(coarse alignment, tilt-compensated heading, specific-force model.)*
9. **D. Titterton, J. Weston**, *Strapdown Inertial Navigation Technology*, 2nd ed., IET, 2004. *(Euler kinematics, sensor error models, coordinated-turn specific force.)*
10. **M. D. Shuster, S. D. Oh**, "Three-Axis Attitude Determination from Vector Observations," *J. Guidance & Control*, 1981. *(TRIAD lineage.)*
11. **J. Diebel**, "Representing Attitude: Euler Angles, Unit Quaternions, and Rotation Vectors," Stanford, 2006. *(quaternion↔Euler.)*
12. **M. S. Longuet-Higgins**, "On the statistical distribution of the heights of sea waves," *J. Marine Res.*, 1952. *($H_s = 4\sqrt{m_0}$.)*
13. **S. O. Rice**, "Mathematical Analysis of Random Noise," *Bell Syst. Tech. J.*, 1944–45. *(zero-crossing rate / spectral-moment period.)*
14. **M. J. Tucker, E. G. Pitt**, *Waves in Ocean Engineering*, Elsevier, 2001. *(spectral moments, sea-state parameters.)* **[verify]**
15. **A. R. J. M. Lloyd**, *Seakeeping: Ship Behaviour in Rough Weather*, Ellis Horwood, 1989. *(roll period, significant single amplitude.)* **[verify]**
16. **D. H. D. West**, "Updating mean and variance estimates: an improved method," *Comm. ACM*, 1979; **T. Finch**, "Incremental calculation of weighted mean and variance," 2009. *(EW mean/variance recursion.)*
17. **A. V. Oppenheim, R. W. Schafer**, *Discrete-Time Signal Processing*, 3rd ed., 2010. *(leaky integrator, first-order high-pass.)*
18. **D. W. Allan**, "Statistics of Atomic Frequency Standards," *Proc. IEEE*, 1966; **W. J. Riley**, *Handbook of Frequency Stability Analysis*, NIST SP 1065, 2008. *(overlapping Allan deviation.)*
19. **IEEE Std 952-2020** (and 952-1997), *Standard Specification Format Guide and Test Procedure for Single-Axis Interferometric Fiber Optic Gyros*, Annex C. *(noise density, bias instability, factor 0.664 — cited in `cal_math.c`.)* **[verify edition]**
20. **A. W. Fitzgibbon, M. Pilu, R. B. Fisher**, "Direct Least Square Fitting of Ellipses," *IEEE TPAMI*, 1999. *(algebraic conic fit.)*
21. **V. Renaudin, M. H. Afzal, G. Lachapelle**, "Complete Triaxis Magnetometer Calibration in the Magnetic Domain," *J. Sensors*, 2010. *(hard/soft-iron sphere & ellipsoid fits.)*
22. **I. D. Coope**, "Circle fitting by linear and nonlinear least squares," *J. Optim. Theory Appl.*, 1993. *(algebraic sphere fit.)* **[verify]**
23. **G. H. Golub, C. F. Van Loan**, *Matrix Computations*, 4th ed., 2013. *(Gaussian elimination with partial pivoting.)*
24. **A. Chulliat et al.**, *The US/UK World Magnetic Model for 2025–2030*, NOAA Technical Report, NCEI/BGS, 2024. *(defining WMM document.)* **[verify exact title/authors of the 2025 report]**
25. **R. A. Langel**, "The Main Field," in *Geomagnetism* (J. A. Jacobs, ed.), Academic Press, 1987. *(Schmidt-normalized ALF recursion — cited in `wmm.c` via WMM Technical Note 28.)*
26. **NIMA TR8350.2**, *Department of Defense World Geodetic System 1984*, 3rd ed. *(WGS-84 ellipsoid constants.)*

27. **N. J. Higham**, *Accuracy and Stability of Numerical Algorithms*, 2nd ed., SIAM, 2002. *(Cancellation in expressions of the form $1 - e^{-x}$, and why an accumulator's width is set by the smallest increment it must still register rather than by the precision of its output.)* [**verify**] — section numbers not checked against a copy.

---

*This document reflects the code at the release it ships with. When a numerical routine changes, update the corresponding section and its **As-implemented** note in the same commit.*